Turbo Coded Space-Time Unitary Matrix Differential Modulation

CCIT Report #365

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Abstract

Non-coherent communication over the Rayleigh flat fading channel with multiple transmit and receive antenna is investigated. Codes achieving bit error rate (BER) lower than 10^{-4} at bit energy over the noise spectral density ratio (E_b/N_0) of 0.8 to 2.8 dB from the capacity limit were found with coding rates of 0.5 to 2.25 bits per channel use. The codes are serial concatenation of turbo code and a unitary matrix differential modulation code. The receiver is based on a high performance joint iterative decoding of the turbo code and the modulation code. Information theoretic arguments are harnessed to form guidelines for code design and to evaluate performance of the iterative decoder.

Keywords

Unitary matrix, multi-antenna, capacity limit, space-time, turbo-code, differential modulation, joint iterative decoding.

I. Introduction

FADING channels are commonly used in wireless communications, see [1] and references therein. When line of sight is absent the Rayleigh distribution is adopted in most cases to model the fading statistics. Recently, there has been considerable interest in the non-coherent multiple antenna communication, where fading coefficients which remain constant for the coherence interval T (in symbols), are unknown (to the transmitter and receiver), since the assumption that channel is known to the receiver is questionable in rapidly changing mobile environment. A general capacity achieving scheme is presented in [2] and [3], where it has been shown that for large Signal to Noise Ratio (SNR), the capacity of multi-antenna Rayleigh fading channel is approached by unitary space-time block codes, in which the signals transmitted by different antennas have equal energy and are mutually orthogonal in the coherence interval. In [4], a signaling scheme, for the Rayleigh flat-fading channel, of unitary space-time modulation is suggested for high signal-to-noise ratio (SNR), or for $T >> M$ as also concluded in [2], [3].

A modulation code for the Multiple-Input Multiple-Output (MIMO) channel was introduced in [5] and in [6], applying a general approach to differential modulation for the MIMO channel based on group codes. We design our demodulator motivated by the low complexity receiver derived in [5] for the signaling scheme therein. The modulation scheme in [5] is shown to be equivalent to a scalar differentially encoded PSK [7] combined with a non-linear space-time transmission rule [6], [8], [9], simplifying the encoding and decoding operations. We base our selection of constellation on the unitarity constraint as in [5], however we do not impose restrictions such as the group structure [5], [8] in an effort to approximate the capacity favored distribution, that is the isotropical one [2].

The concatenation of turbo code and a modulation code for the MIMO channel was introduced in [10], [11], [12], [13]. In [10] an arbitrarily turbo coded modulation scheme (with a trivial constellation mapper as a modulation code) is presented, assuming known channel fading coefficients, and using a sub-optimal decoding algorithm, offering a significant improvement over the traditional space-time codes [14]. This is extended in [11], where a turbo code is concatenated with a unitary space-time modulator, assuming no channel information, and using a sub-optimal receiver. We compare the performance of [11] to our suggested system in section V.

The concatenated codes are decoded by the iterative algorithm designed originally for turbo-codes in [15] and extended in later works [16], [17], [18] to the general case of interleaved concatenated codes. Further extensions of the iterative algorithm which place the decoding of additional elements specific to a coded communication system inside the iterative decoder were investigated in previous works, for example [19], [20] and references therein. The additional elements are treated similarly to another component code of a concatenated system, yielding decoder structures similar to ours. Some examples of such system specific elements are modulation codes and multiple symbol differential modulation over non-coherent channels [19], [20], joint decoding and channel estimation [21], [22]. In all the given examples a significant improvement has been achieved over the separate detection of system elements and of the outer code.

Motivated by the results of [2], and using the concepts introduced in [6], we investigate a concatenated turbo encoder and modulation code for achieving reliable communication at SNR reasonably close to the capacity limit. The block fading channel models investigated are applicable for mobile wireless systems, TDMA systems and for frequency hopping systems.

In section II we present the system model, including the channel model, the transmitter structure, and the receiver. In section III we outline the guidelines for code selection and evaluation. In section IV we extend the observation block of the receiver for better utilization of time dependency. Section V includes simulation results and performance analysis. Conclusions in section VI terminate the paper.

II. System Model

A. Channel Model

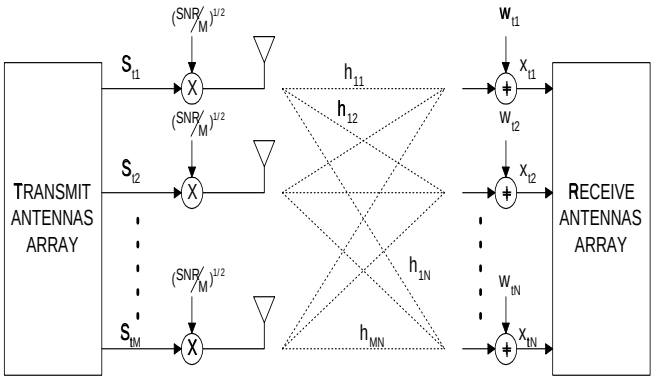


Fig. 1. Block Diagram of a multiple antenna fading channel.

A channel model comprising M transmit and N receive antennas may be described by its baseband discrete time equivalent as follows

$$X = \sqrt{\frac{\rho}{M}} \cdot H \cdot S + W \tag{1}$$

where S is a MxT matrix of transmitted signals, and s_{mt} is the signal transmitted from antenna m at time t. H is a $N \times M$ channel matrix. Its elements h_{nm} are independent complex Gaussian distributed, with mean zero and a unit variance. During a coherence block T, the channel matrix H is fixed, and unknown to the transmitter and the receiver. W is a $N \times T$ additive noise matrix, with independent $CN(0,1)$ distributed elements. ρ is the SNR at each receive antenna, while $E[|s_{mt}|^2] = 1$.

The investigated fading channel models are:

1. *Non-overlap* - Successive coherence intervals are statistically independent.
2. *Overlap* - Fading is relatively slow. The independent fading coefficients are random complex Gaussian processes with a discrete auto-correlation function

$$R(k) = e^{-k \cdot T_s/\tau} \tag{2}$$

where k is a discrete time index, T_s is the sampling period, and τ is chosen so that during a transmitted block, in every coherence interval T, variation of fading coefficients is small, approximating the fixed channel assumption.

In both scenarios $T = 2M$ is used. It is known from [3] that for $T/2 \geq \min\{M, N\}$ and large ρ the capacity achieving S has orthonormal rows and a unitary isotropic distribution. From [2] and [3] it is clear that capacity of multi-antenna channel can be approached for large T or ρ by code matrices with equal energy, isotropically distributed orthogonal rows. Accordingly, our attention was drawn to codes with the property

$$S \cdot S' = T \cdot I \tag{3}$$

where S is an $M \times T$ *unitary code matrix*. The rows of S are kept as constraint free as possible, to achieve distribution similar to the isotropical one, thus we do not impose restrictions, as for example in the group structure introduced in [5], [6].

B. Transmitter

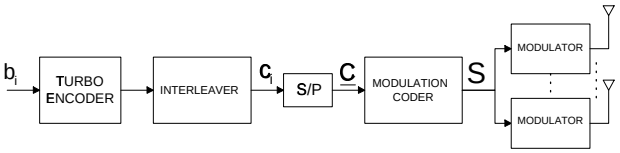


Fig. 2. Block diagram of a transmitter with turbo encoder and unitary matrix differential modulation coder.

A block diagram of the transmitter, including a turbo encoder and a modulation encoder is presented in Figure 2. The input is a N_{inf} long stream of bits b_i. This bit-stream is encoded by a standard turbo encoder, which includes a parallel concatenation two convolutional encoders as in [15]. The turbo-encoded bits are bitwise interleaved and parsed into words $\underline{C}$. Each word $\underline{C}$ is mapped into a $M \times M$ *unitary code matrix* D_i.

In the *Non-Overlap* case, a unitary matrix differential modulation is achieved by choosing an arbitrary unitary $M \times M$ reference matrix A_i. A transmitted matrix S_i consists of A_i as a reference followed by a differential modulated matrix $A_i \cdot D_i$. It is clear that all possible $A_i \cdot D_i$ are unitary. Hence the transmitted matrix S_i is given by

$$S_i = \sqrt{M} \cdot [\ A_i \quad A_i \cdot D_i\ ]_{M \times 2M} \qquad i = 1, 2, ..., N_{block} \tag{4}$$

It is easy to show [6] that S_i conforms to the orthogonality property (3). D_i should approximate unitary uniform distribution, which is the optimal distribution according to [23]. Note that this differential structure does not use all the possible degrees of freedom inherent in $M \times 2M$ matrices [3] over a non-coherent channel. In the *Overlap* case the structure defined by (4) is retained while A_i is the previous differential modulated unitary $M \times M$ matrix transmitted as in [6]. Then the transmitted $M \times M$ matrix for the *Overlap* case is given by

$$S_0 = \sqrt{M} \cdot A_0$$
$$S_i = [S_{i-1} \cdot D_{i-1}]_{M \times M} \quad i = 1..N_{block} - 1 \tag{5}$$

C. Receiver

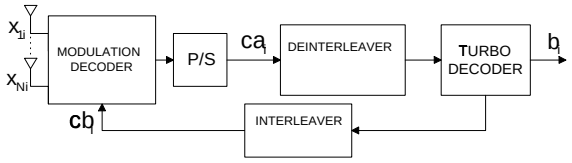


Fig. 3. Block diagram of iterative receiver.

C.1 General Structure

The block diagram of the receiver, including a demodulation decoder and a turbo decoder is presented in Figure 3, the notations of which are used in the following.

The receiver approximates the optimal joint detection of the turbo code and modulation code by means of iterative processing. Using principles presented for example in [19], the modulation decoder (MD) operating on the words $\underline{C}$ of the coded bits, first computes soft metrics $ca_i = P(c_i = 1|X, cb_i)$ of the turbo encoded bits c_i, based on the received $M \times T$ matrices (X_{i-1}, X_i) and the extrinsic side information $cb_i = P(c_i = 1)$. The soft metrics are de-interleaved bitwise and decoded by one iteration of a turbo decoder [15] which also provides the soft metrics of the coded bits cb_i used by the modulation decoder in the next iteration. It is based on Serial In Serial Out (SISO) modules as presented in [16], and also used in [19]. It includes two component code decoders, each decoding one of the component recursive convolutional codes by means of Bahl et al. algorithm [24]. This process is repeated on each received turbo-coded block of N_{inf} bits for N_{it} iterations.

C.2 Modulation Decoder

First, the joint probability $M(\underline{C}) \doteq P(X, \underline{C})$ is computed for all possible codewords $\underline{C}$, based on the side information $cb_i = P(c_i = 1)$ and received $M \times T$ matrix (X_{i-1}, X_i)

$$M(\underline{C}) = P(X_{i-1}, X_i|D(\underline{C})) \cdot \prod_{j=1}^{N_c} P(c_j(\underline{C})|cb_j) \tag{6}$$

where N_c is the bit-length of each codeword $\underline{C}$, and $P(c_i(\underline{C})|cb_i) = c_i \cdot cb_i + (1 - c_i) \cdot (1 - cb_i)$. The conditional pdf of the received matrices (X_{i-1}, X_i) corresponding to the two transmitted matrices in 4, can be derived similarly to eq. (7), (8) in [5] as

$$P(X_{i-1}, X_i|\underline{C}) = const \cdot \exp^{\frac{2\rho}{1+T\rho/M} \cdot real\{tr[X_{i-1}D(\underline{C})X_i^\dagger]\}} \tag{7}$$

where *const* contains irrelevant arguments of $P(X_{i-1}, X_i|\underline{C})$ which are independent on $D(\underline{C})$, and $D(\underline{C})$ is the unitary code matrix mapped by the codeword $\underline{C}$, X_i are the received matrices corresponding to the square transmitted matrices comprising of S_i in (4), (5). The final bit metrics ca_i is calculated as in eq. (10), (11) of [19], marginalizing the probability of code words $P(X, \underline{C})$ and extracting extrinsic information.

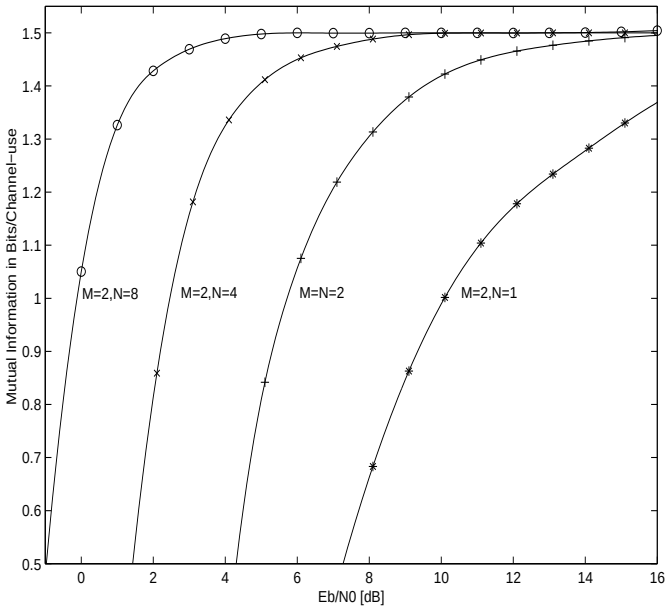


Fig. 4. Word AMI calculation in the *non-overlap* scheme, for $T = 4$, two transmit antennas and various number of receive antennas, using unitary code matrices, and matrix differential modulation.

III. Code Evaluation Guidelines

Since the modulation code inputs are words $\underline{C}$ of turbo coded and interleaved bits, all input words are practically equiprobable and statistically independent. Thus to operate near capacity a modulation code must achieve Average Mutual Information (AMI) between the input words $\underline{C}$ and the channel output statistics X close to capacity with equiprobable independent input words $\underline{C}$. Thus maximizing $I(\underline{C};X)/T$, which is denoted as the word AMI, where $I(x;y) = E\log_2\frac{P(x|y)}{P(x)}$ as in [25], happens to be an essential criterion in modulation code selection.

In the *Non-Overlap* scheme the word AMI is also the code restricted channel capacity, due to the strict independence between consecutive blocks. In the *Overlap* scheme the correlated fading channel as introduced in (2) implies dependency between consecutive blocks, rendering capacity calculation rather complex, however our receiver does not utilize this dependence. Furthermore the dependence is dispersed by the interleaver, thus for long blocks the word AMI is equivalent to capacity as a performance bound, taking the limitations of the sub-optimal receiver into account. The latter implies that full channel capacity is necessarily an upper bound for the word AMI.

Figure 4 presents a calculation result for word-AMI, in the *non-overlap* scheme, for $T = 4$, two transmit antennas and various number of receive antennas. The transmit matrices are consrtucted as in eq. (4), where D_i is a 2×2 matrix containing QPSK elements. The mapping of 6-bit word $\underline{C}$ to unitary matrix D_i required in eq. (4) is performed by mapping three pairs of bits to three QPSK symbols by Gray code, determining the fourth symbol by the unitary constraint, and thus using unitary code matrices, and matrix differential modulation eq. (4). It is noticeable from figure 4 that for $M = 2$ transmit antennas, the greatest capacity improvement, in means of E_b/N_0, is achieved when multiplying the number of receive antennas from $N = 1$ to $N = 2$. When increasing the receive antenna diversity to $N = 4$ and $N = 8$ the capacity gain is still significant, even though the coherence interval is $T = 4$.

The modulation decoder uses extrinsic information received from the turbo decoder to improve the

input probability $ca_i = P(c_i = 1|X)$ for the turbo decoder. At first iteration the side information cb_i is missing, thus the best the modulation decoder can do is produce $P(c_i = 1|X)$, the information content of which is limited by $I(c_i; X)$, denoted in the following as bit AMI, and clearly it is upper bounded by the word AMI, thus the initial conditions of the iterative decoder are limited by bit AMI. The relevance of bit AMI to performance of an iterative decoder was also demonstrated in [19] and in [26]. Thus bit AMI is an additional yet empiric criterion in modulation code selection. Clearly, performance with modulation feedback is upper bounded by word AMI, and without modulation feedback by bit-AMI.

IV. Extended Observation Interval

Two approaches are demonstrated for a better utilization of the channel characteristics for increasing the code restricted capacity. Both approaches achieve the same goal by extending the observation interval at the receiver, and performing sub-optimal decoding, which considers the correlation in the fading random process.

The first approach is denoted as the Reference Differential Encoding (RDE). In this method a single reference matrix is used, for differential encoding over a block of k transmitted matrices, i.e. $k-1$ data carrying matrices are differentially modulated using a common arbitrarily chosen unitary reference matrix. A sub-optimal decoder, which keeps the decoding complexity constant for any k, is introduced hereby.

The second approach is based on the standard differential encoder (SDE) as introduced in section II-B. A sub-optimal decoder, which keeps the decoding complexity constant for any k is suggested. The modulation decoder operates on k received matrices performing joint detection.

A. RDE Based System

A unitary matrix differential modulation is achieved by choosing an arbitrary unitary $M \times M$ reference matrix A_i. A transmitted matrix S_i consists of A_i as a reference followed by a differential modulated matrix $A_i \cdot D_{i,1}$, and $A_i \cdot D_{i,2}$, up-to $A_i \cdot D_{i,k-1}$. It is clear that all possible $A_i \cdot D_{i,j}$ are unitary. Hence the $M \times kM$ transmitted matrix S_i is given by

$$S_i = \sqrt{M} \cdot \left[\begin{array}{cccc} A_i & A_i \cdot D_{i,1} & A_i \cdot D_{i,2} & \ldots \quad A_i \cdot D_{i,k-1} \end{array} \right] \tag{8}$$

where $D_{i,j}$ is the data carrying matrix and A_i is the reference matrix. The fading coefficients are assumed to be fixed during the transmission of S_i, which hereby defines a coherence interval of $T = k \cdot M$.

The received signal matrix may then be described as a block matrix

$$X_i = \left[\begin{array}{cccc} X_{i,0} & X_{i,1} & \ldots & X_{i,k-1} \end{array} \right] \tag{9}$$

The receiver has to compute $P(X_i|\underline{C})$ for all possible codewords $\underline{C}$. This conditional pdf can be derived similarly to eq. (7), (8) in [5] as

$$P(X_i|\underline{C}) = const \cdot \exp^{\frac{2\rho}{1+T\rho} \cdot real\{tr[X_i S_i^\dagger S_i X_i^\dagger]\}} \tag{10}$$

where *const* contains irrelevant arguments of $P(X_i|\underline{C})$ which are independent on S_i. It is clear from eq. (10) that the main computation task is computing

$$\begin{array}{rl} weight_i & = real\{tr[X_i S_i^\dagger S_i X_i^\dagger\} \\ & = real\{tr[S_i^\dagger S_i X_i^\dagger X_i]\} \end{array} \tag{11}$$

for all possible $\underline{C}$. Rotating the matrix multiplication order in eq. (11) is allowed because the trace is invariant to rotational reordering in multiplication of matrices. The matrix multiplication in eq. 11 may be described in the sub-matrices form, according to the definition of S_i in eq. (8) and X_i in eq. (9). After multiplying $S_i^\dagger S_i$, and $X_i^\dagger X_i$, and omitting the discrete time index i for compactness of the representation, *weight* is given by the $real\{tr[]\}$ of

$$[S^\dagger S]\cdot[X^\dagger X] = \begin{bmatrix} A^\dagger A & A^\dagger A D_1 & \cdots & A^\dagger A D_{k-1} \\ D_1^\dagger A^\dagger A & D_1^\dagger A^\dagger A D_1 & \cdots & D_1^\dagger A^\dagger A D_{k-1} \\ \vdots & \vdots & \ddots & \vdots \\ D_{k-1}^\dagger A^\dagger A & D_{k-1}^\dagger A^\dagger A D_1 & \cdots & D_{k-1}^\dagger A^\dagger A D_{k-1} \end{bmatrix} \cdot \begin{bmatrix} X_0^\dagger X_0 & X_0^\dagger X_1 & \cdots & X_0^\dagger X_{k-1} \\ X_1^\dagger X_0 & X_1^\dagger X_1 & \cdots & X_1^\dagger X_{k-1} \\ \vdots & \vdots & \ddots & \vdots \\ X_{k-1}^\dagger X_0 & X_{k-1}^\dagger X_1 & \cdots & X_{k-1}^\dagger X_{k-1} \end{bmatrix} \tag{12}$$

The matrix multiplication result can be further simplified by using the unitarity property eq. (3), and by dropping the additive elements dependent only on X_i. Then *weight* is given by

$$\begin{aligned} weight = \; & const + 2real\{tr[\textstyle\sum_{i=1}^{k-1} D_i X_i^\dagger X_0 + \sum_{i=2}^{k-1} D_1^\dagger D_i X_i^\dagger X_1 + \\ & + \textstyle\sum_{i=3}^{k-1} D_2^\dagger D_i X_i^\dagger X_2 + \cdots + D_{k-2}^\dagger D_{k-1} X_{k-1}^\dagger X_{k-2}]\} \end{aligned} \tag{13}$$

where *const* represents the X_i dependent elements. Since the computation complexity grows exponentially with codeword length, and *weight* is calculated for an equivalent codeword of length $k \cdot length(\underline{C})$, the reduction of the decoding complexity is essential.

In order to turn the decoding process into a realistic task it is suggested to compute *weight* for each data carrying matrix D_i $(0 \le i \le k-1)$, using the estimation of all the other $\{D_j\}_{j=0:j\neq i}^{j=k-1}$ instead of their real value. The expected value $\{E[D_j]\}_{j=0:j\neq i}^{j=k-1}$ can be directly calculated from the extrinsic information received from the turbo decoder at each iteration, as shown later in eq. (16).

Thus the expression of *weight* may be approximated by $weight_{est}$, and further simplified by using the estimations of some of codewords, and by dropping the irrelevant elements for decoding, which are the additive elements dependent only on X or $E[D_j]$. Hence *weight* is given by

$$\begin{aligned} weight_{est,k-1} = const_1 + 2real\{tr[D_{k-1}X_{k-1}^\dagger X_0 + E[D_1^\dagger]D_{k-1}X_{k-1}^\dagger X_1 + \cdots + \\ + E[D_{k-2}^\dagger]D_{k-1}X_{k-1}^\dagger X_{k-2}]\} \end{aligned}$$

$$\Downarrow$$

$$weight_{est,k-1} = const_2 + 2real\{tr[D_{k-1}X_{k-1}^\dagger \cdot (X_0 + \sum_{i=1}^{k-2} X_i E[D_i^\dagger])]\} \tag{14}$$

where $const_1$ includes elements dependent on X only, and $const_2$ includes also elements dependent on the combination of X and $E[D_j]$. Eq. (14) represents the sub-optimal expression for *weight* required for computing the probability of the codeword $\underline{C}$, which depends only on D_{k-1} $(D_{k-1}(\underline{C}))$. The process is repeated for each $\{D_j\}_{j=0}^{j=k-1}$, and thus allows computation of approximated probabilities

for codewords of length $length(\underline{C})$ instead of $k \cdot length(\underline{C})$. The generalized expression of $weight_{est,j}$ for $(1 \leq j \leq k-1)$ is given by

$$weight_{est,j} = const_2 + 2real\{tr[D_j X_j^\dagger \cdot (X_0 + \sum_{i=1, i\neq j}^{k-1} X_i E[D_i^\dagger])]\} \tag{15}$$

The estimation of D_i is given by the optimal mean square error (MSE) estimator. Given the codewords probabilities computed from the turbo decoder extrinsic information, see also (6), the estimated data matrices are

$$E[D_i] = \sum_{\underline{C}} P(\underline{C}|\underline{CB}) D(\underline{C}) \tag{16}$$

where $\underline{CB}$ is the vector of the coded bits probability, received from the turbo decoder, and the expression $X_0 + \sum_{i=1, i\neq j}^{k-1} X_i E[D_i^\dagger]$ replaces X_0 as an improved reference matrix. Clearly, the *non-overlap* case, eq. (4) and the corresponding receiver eq. (7), are identical to the presented RDE system eq. (8), (15), when the observation interval is $T = 2M$.

A further approximation that was found beneficial for improved simulation performance is to give different weighting for the estimated reference, than the received reference in eq. (15). When $weight_{est,j}$ is given by

$$weight_{\alpha_{est},j} = const_2 + 2real\{tr[D_j X_j^\dagger \cdot (X_0 + \alpha \cdot \sum_{i=1, i\neq j}^{k-1} X_i E[D_i^\dagger])]\} \tag{17}$$

where α is the empirical weighting factor of the estimated reference. The best simulation performance was achieved for $\alpha = 1/(K-1)$.

B. SDE Based System

This system uses standard differential encoding for matrix differential modulation, and uses extended observation interval in the receiver. It can be used in the *overlap* and *non-overlap* cases. A reduced complexity sub-optimal decoder is derived here.

The sub-block i of successive k transmit matrices, in the *non-overlap* case, is given by

$$S_i = \sqrt{M} \cdot [\ A_i \quad A_i \cdot D_{i,1} \quad A_i \cdot D_{i,1} \cdot D_{i,2} \quad \dots \quad A_i \cdot \textstyle\prod_{j=1}^{k-1} D_{i,j}\ ] \tag{18}$$

where $D_{i,j}$ is the data carrying matrix and A_i is the reference matrix. All $D_{i,j}$ and A_i are $M \times M$ unitary square matrices. The main difference from eq. (4) is that instead of a single data carrying matrix $(D_{i,j})$ after the reference matrix $k-1$ data matrices are transmitted. The fading coefficients are assumed to be fixed during the transmission of S_i as in section IV-A.

In the *overlap* case the transmit matrix is given by eq. (5).

For deriving a reduced complexity receiver let us assume that the transmitted block, omitting the time index i is given by

$$S = \left[\ S_0 \quad S_0 D_1 \quad S_1 D_2 \quad \dots \quad S_{k-1} D_{k-1}\ \right] \tag{19}$$

where S_j is a unitary $M \times M$ transmit matrix given by the elements of the transmit matrix specified in eq. (18). The received matrix may thus be described by

$$X = \left[\ X_0 \quad X_1 \quad \dots \quad X_{k-1}\ \right] \tag{20}$$

where each element in the received matrix is an $M \times M$ received matrix. We need to compute here $weight_i$ as specified by eq. (11). when multiplying $S_i^\dagger S_i$, and $X_i^\dagger X_i$, and omitting the discrete time index i for compactness of the representation, $weight$ is given by the $real\{tr[]\}$ of

$$[S^\dagger S]\cdot[X^\dagger X] = \begin{bmatrix} S_0^\dagger S_0 & S_0^\dagger S_0 D_1 & \cdots & S_0^\dagger S_0 \prod_{j=1}^{j=k-1} D_j \\ D_1^\dagger S_0^\dagger S_0 & D_1^\dagger S_0^\dagger S_0 D_1 & \cdots & D_1^\dagger S_0^\dagger S_0 \prod_{j=1}^{j=k-1} D_j \\ \vdots & \vdots & \ddots & \vdots \\ \prod_{j=k-1}^{j=1} D_j^\dagger S_0^\dagger S_0 & \cdots & \cdots & \prod_{j=k-1}^{j=1} D_j^\dagger S_0^\dagger S_0 \prod_{j=1}^{j=k-1} D_j \end{bmatrix} \cdot \begin{bmatrix} X_0^\dagger X_0 & X_0^\dagger X_1 & \cdots & X_0^\dagger X_{k-1} \\ X_1^\dagger X_0 & X_1^\dagger X_1 & \cdots & X_1^\dagger X_{k-1} \\ \vdots & \vdots & \ddots & \vdots \\ X_{k-1}^\dagger X_0 & X_{k-1}^\dagger X_1 & \cdots & X_{k-1}^\dagger X_{k-1} \end{bmatrix} \tag{21}$$

The matrix multiplication result can be further simplified by using the unitarity property eq. (3), thus eq. (21) is given by

$$[S^\dagger S]\cdot[X^\dagger X] = M\cdot \begin{bmatrix} I & D_1 & \cdots & \prod_{j=1}^{j=k-1} D_j \\ D_1^\dagger & I & \cdots & \prod_{j=2}^{j=k-1} D_j \\ \vdots & \vdots & \ddots & \vdots \\ \prod_{j=k-1}^{j=1} D_j^\dagger & \prod_{j=k-1}^{j=2} D_j^\dagger & \cdots & I \end{bmatrix} \cdot \begin{bmatrix} X_0^\dagger X_0 & X_0^\dagger X_1 & \cdots & X_0^\dagger X_{k-1} \\ X_1^\dagger X_0 & X_1^\dagger X_1 & \cdots & X_1^\dagger X_{k-1} \\ \vdots & \vdots & \ddots & \vdots \\ X_{k-1}^\dagger X_0 & X_{k-1}^\dagger X_1 & \cdots & X_{k-1}^\dagger X_{k-1} \end{bmatrix} \tag{22}$$

The matrix multiplication result can be further simplified by dropping the additive elements dependent only on X_i, $weight$ is given by

$$\begin{aligned} weight = \; & const + 2real\{tr[\textstyle\sum_{i=1}^{k-1}\prod_{j=1}^{j=i} D_j X_i^\dagger X_0 + \sum_{i=2}^{k-1}\prod_{j=2}^{j=i} D_j X_i^\dagger X_1 + \cdots \\ & + D_{k-1} X_{k-1}^\dagger X_{k-2}]\} = \\ = \; & const + 2real\{tr\left[\textstyle\sum_{m=0,n=1}^{k-2,k-1}\prod_{j=m+1}^{n} D_j X_n^\dagger X_m\right]\} \end{aligned} \tag{23}$$

In order to turn the decoding process into a realistic task it is suggested to compute $weight$, similarly to RDE (section IV-B), for each data carrying matrix D_i $(0 \le i \le k-1)$, using the estimation of all the other $\{D_j\}_{j=0:j\neq i}^{j=k-1}$ instead of their real value. The expected value $\{E[D_j]\}_{j=0:j\neq i}^{j=k-1}$ can be directly calculated from the extrinsic information received from the turbo decoder at each iteration, eq. (16). Thus the expression of $weight$ may be approximated by $weight_{SDE_{est}}$, and further simplified by using the estimations of some of codewords, and by dropping the irrelevant elements for decoding from eq. (23). Decoding D_i implies then, on the following $weight$

$$\begin{aligned} weight_{D_i,SDE_{est}} = \; & const_1 + 2real\{tr[\textstyle\sum_{m=0,n=i}^{i-1,k-1}\prod_{m+1\le j_1\le n} E[D_{j_1}]\cdot \\ & \cdot D_i \cdot \prod_{i+1\le j_2\le n} E[D_{j_2}] X_n^\dagger X_m]\} \end{aligned} \tag{24}$$

It should be noted that for $m{+}1 > n$ $\prod_{j=m+1}^{j=n} E[D_j]$ is replaced by I. In the *overlap* case, $weightD_i, SDE_{est}$ can be calculated for the same i, i.e. the k-observation block may be shifted by one (a sliding window)

for every D_i, thus eliminating the boundary effect of computing D_{k-1} using estimations of previous received matrices *only*.

An example for the *overlap* case, with $k = 4$, and $i = 2$ is chosen to demonstrate how the decoding operation from eq. (24) simplifies. Eq. (24) gets the following form

$$\begin{aligned} weight_{D_2,SDE_{(est)}} = \quad & 2real\{tr[(D_2X_2^\dagger X_1 + E[D_1]D_2X_2^\dagger X_0 + \\ & + D_2E[D_3]X_3^\dagger X_1 + E[D_1]D_2E[D_3]X_3^\dagger X_0)]\} \end{aligned} \tag{25}$$

including only 4 elements for summation.

Clearly, the decoder given by eq. (23) is identical to eq. (7) when the observation interval is $T = 2M$.

A further modification was found beneficial for improved simulation performance, like in the RDE, that is to give different weighting for the estimated product $X_1\dagger X_2$, than the received product in eq. (25). This results in the following

$$\begin{aligned} weight_{D_2,SDE_{(est)}} = \quad & 2real\{tr[D_2X_2^\dagger X_1 + \alpha(E[D_1]D_2X_2^\dagger X_0 + \\ & + D_2E[D_3]X_3^\dagger X_1) + \beta(E[D_1]D_2E[D_3]X_3^\dagger X_0)]\} \end{aligned} \tag{26}$$

where α is the empirical weighting factor of the estimated reference. The best simulation performance was achieved for $\alpha = 1/(K-1)$, $\beta = 0$.

V. Simulation, Performance and Analysis

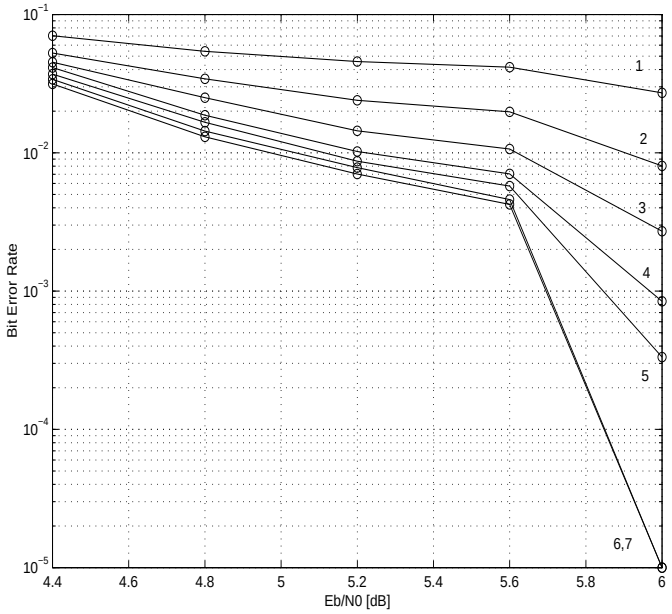


Fig. 5. Performance of the iterative receiver for 7 iterations, Coding Rate of 1.125, with overlapping, $\tau = 128$ from (2), $M = N = 2$

We present the performance of the turbo-coded unitary matrix differential modulation for two transmit and two or four receive antennas, compared to capacity limit in Figure 6, 7 and Figure 8. In both figures the Bit Error Rate (BER) is measured for the bit energy over the noise spectral density E_b/N_0, which is defined as

$$E_b/N_0 \doteq \frac{SNR}{R_s} \equiv \frac{\rho}{R_s} \tag{27}$$

where R_s is the system rate given in uncoded information bits per channel use.

Component codes of the turbo code are recursive systematic convolutional codes, described by $(g_0, g_1) = (5, 7)$, where g_0 and g_1 are feed-forward and feedback generating polynomials. The turbo-code employs a uniform pseudo random interleaver [17]. The performance was evaluated for 3 different turbo-code rates $R_c = \frac{1}{3}, \frac{1}{2}, \frac{3}{4}$. The interleaver of the turbo-coded bits is also a uniform random interleaver over the transmitted block, both interleavers are selected randomly for each simulation block. The block size is $N_{inf} = 3,000$. For $R_c = \frac{1}{2}$ a block of $N_{inf} = 15,000$ was simulated as well. The iterative decoding included $N_{it} = 10$ iterations.

Two transmit antennas, along with two or four receive antennas are used, with a QPSK symbols at each transmit antenna. The mapping of 6-bit word $\underline{C}$ to unitary matrix D_i required in eq. (4) is performed by mapping three pairs of bits to three QPSK symbols by Gray code. The 3-QPSK symbols are three elements of the 2×2 matrices D_i and the fourth symbol is determined by the unitarity constraint. Thus, the modulation rate is 1.5 and 3 coded bits per channel sample for the *non-overlap* and the *overlap* scheme respectively. The system rates corresponding to the turbo-code rates are $R_{s(NoOverlap)} = \frac{1}{2}, \frac{3}{4}, \frac{9}{8}$ [bits / channel use] for the *non-overlap* scheme, and $R_{s(Overlap)} = 1, \frac{3}{2}, \frac{9}{4}$ [bits / channel use] for the *overlap* scheme. The per-symbol-Gray mapping described above is known to maximize the bit AMI for a single antenna coherent communication [27], and achieved the best bit AMI for our MIMO case after examination of a few available mappings.

Figure 5 demonstrates a typical performance of the iterative receiver in the *overlap* scheme, with $R_c = \frac{3}{4}$, $R_s = \frac{9}{4}$, $N_{inf} = 3000$, with a modulation feedback, the performance resembles qualitatively that of a standard turbo decoder [15], improving over iterations and exhibiting a steep BER versus E_b/N_0 slope.

A. Non-Overlap Scheme

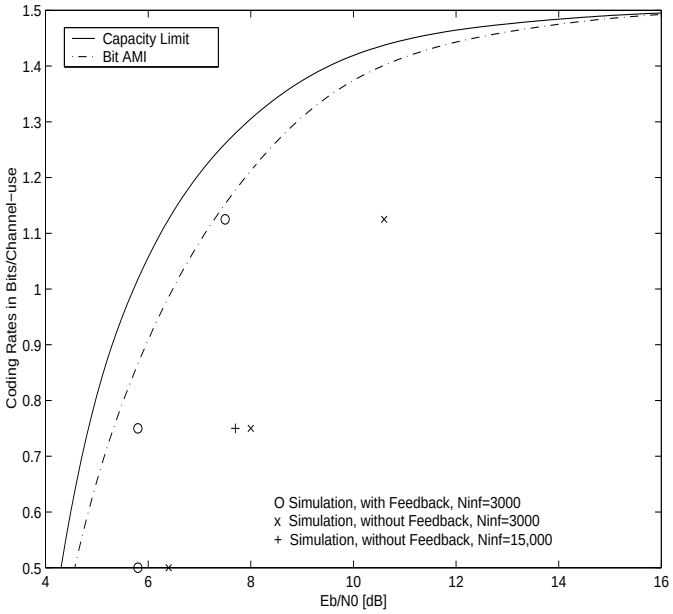


Fig. 6. *Non-Overlap* scheme. Simulation results for BER=10^{-4}, compared to word AMI, and bitwise-AMI. $M = N = 2$.

In Figure 6, we evaluate the performance of the described system in the *non-overlap* scheme demonstrating the efficiency of the turbo code and the sub-optimal joint iterative decoder. It can be seen

that the system operating points at which BER is less than 10^{-4} are only 0.8-1.4 dB from the code constrained channel capacity. The upper bound on the simulated system performance is the word AMI when exploiting the extrinsic information of the turbo-decoder, i.e. when using the modulation feedback (see section III), and the bit AMI bound in a system without modulation feedback.

The contribution of the iterative joint demodulator and turbo decoder is larger than predicted by the gap between the bit AMI and the word AMI, as clearly seen from Figure 6, where operating points at which BER is less than 10^{-4}, are 1.7-3.2 dB from bit AMI bound without modulation feedback. Increasing the block length of the data from $N_{inf} = 3000$ to $N_{inf} = 15,000$ yield 0.3 dB improvement, suggesting that for longer blocks the performance may get close to bit AMI bound.

A similar phenomenon is evident in [28], where the minimum required SNR resulting from random coding exponent as defined by Gallager [25], which in turn leads to a word error rate lower than a given threshold, is computed. It is shown by means of error exponent, that SNR required to achieve a given error rate increases with decreasing block-length faster for bitwise decoding (see Fig. 1, BICM in [28]), than for wordwise decoding (see Fig. 1, MLD in [28]). This observation is in agreement with our results, since the iterative receiver attempts to approximate optimal decoding.

B. Overlap Scheme

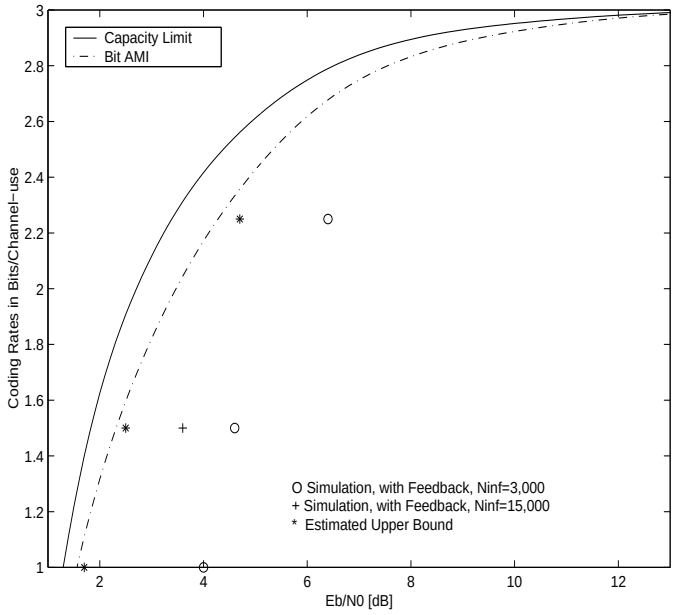


Fig. 7. *Overlap* scheme. Simulation results for BER=10^{-4}, compared to word AMI, and bitwise-AMI. $M = N = 2$.

In Figures 7 and 8, the performance of the system in the *overlap* scheme is evaluated. It should be noted that the channel capacity is calculated for a hypothetical channel, such that the fading coefficients are fixed for two consecutive matrix transmissions, and independent on any previous transmission, yielding an over-estimation of capacity. However, the channel was characterized more realistically for simulation by the correlation function (2).

A first order estimation of the SNR loss, due to the variance of the fading coefficient in a single transmitted block was done. For clarity we analyze the one-dimensional case, the same results can be derived for the multi-dimensional case. It can be shown that the channel correlation function (2) determines the discrete random process of the fading as follows

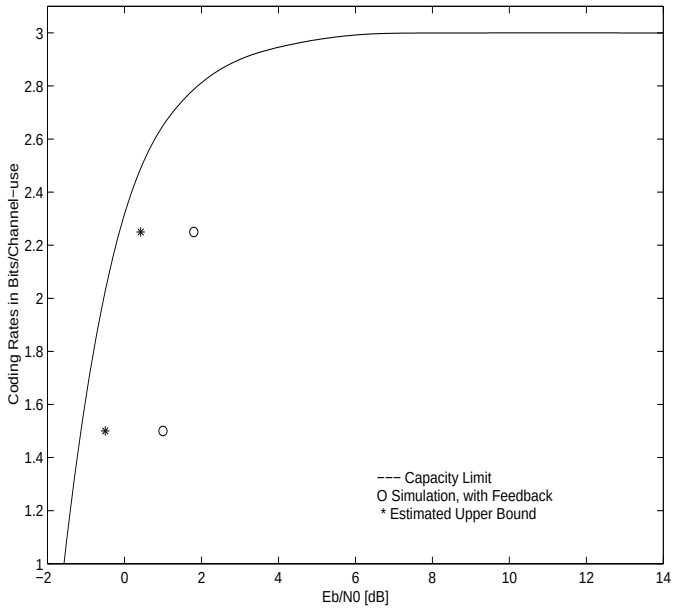


Fig. 8. *Overlap* scheme. Simulation results for BER=10^{-4}, compared to word AMI, $N_{inf} = 3,000$, $M = 2, N = 4$.

$$H_i = (1-\alpha) \cdot H_{i-1} + \sqrt{\alpha \cdot (2-\alpha)} \cdot W_i \tag{28}$$

where $\alpha = 1 - e^{-T_s/\tau}$ (thus $R(k) = (1-\alpha)^k$ as in (2)) and W_i is the additive complex Gaussian noise with independent $CN(0,1)$ distribution. Thus, the received signal may be represented as

$$X_i = (1-\alpha) \cdot \sqrt{\frac{\rho}{M}} \cdot H_{i-1} \cdot S_i + \sqrt{\alpha \cdot (2-\alpha) \cdot \frac{\rho}{M}} \cdot S_i \cdot W_{i,1} + W_{i,2} \tag{29}$$

where S_i is the single-dimensional transmit symbol at time i. H_i is given by (28). $W_{i,1}$, $W_{i,2}$ are the additive complex Gaussian noise with independent $CN(0,1)$ distribution. This representation resembles the AR model representation eq. (10) in [29]. Clearly, the first term in (29) is the main attenuated signal and the second term is additional uncorrelated noise.

The SNR loss $L_{Overlap}$ may now be computed by

$$L_{Overlap} = \frac{1 + \alpha \cdot (2-\alpha) \cdot \rho}{(1-\alpha)^2} \tag{30}$$

Result (30) is denoted in Figure 7 as an estimated upper bound, which is 0.45-1.3 dB below the word AMI bound for the different rates and SNRs. Some of the additional degradation could be caused by the correlation of the fading coefficients between observation intervals reducing the time-diversity of the scheme. Therefore, it is interesting to see the difference in performance between the case of $N = 2$ (figure 7) and $N = 4$ (figure 8) for $N_{inf} = 3,000$. For the latter ($N = 4$), the performance is 1.4-1.5 dB from the estimated upper bound, in contrast to 2.5 dB for $N = 2$, which indicates that the time-diversity was replaced by additional space-diversity. It is also interesting to notice that increasing the block length of the data from $N_{inf} = 3,000$ to $N_{inf} = 15,000$, in the case of $N = 2$, yields a 1 dB better performance, and only a 1.1 dB gap from the estimated upper bound at BER less than 10^{-4}.

System Model	T [Channel Uses]	$E_b/N_0[dB]$ $for BER \leq 10^{-4}$	R_s [Bits per Channel Use]
Non-Overlap	$2M$	5.8	0.5
RDE	$5M$	3.2	0.8
Overlap	$2M$	4	1
SDE	$4M$	3.4	1

TABLE I
SIMULATION RESULTS SUMMARY OF RDE, SDE SCHEMES VERSUS THE *Overlap and Non-Overlap* SCHEMES

C. SDE and RDE Schemes

In Table I, the performance of the system in the RDE scheme is compared with the system in the *non-overlap* scheme, and the system in the SDE scheme is compared with the system in the *overlap* scheme. The channel in the second case was realistically characterized for simulation by the correlation function in (2).

In all four simulations the turbo-code rate chosen is $R_c = \frac{1}{3}$, which is expected to be most robust in means of error correction, hence convey maximum side information for the modulation decoder. In the RDE scheme a $k = 5$ was chosen, i.e. the reference matrix was recalculated every iteration using $4[M \times M]$ transmitted matrices. In the SDE scheme a $k = 3$ was chosen for simulation, thus a reference matrix was recalculated every iteration using a preceding and the next successive transmitted matrices. It may be noticed that in both cases the overall performance gain is 0.6 dB, when the inherent gain is first subtracted (e.g. for the RDE case: $\log 0.5/0.8$).

D. Performance Comparison to Other Systems

In Figure 5 of [11] the simulation performance of a turbo code concatenated with a unitary modulation code is given for 2 transmit and 2 receive antennas, with $T = 5$, $N_{inf} = 3,500$, $R_s = 1.75$, and turbo code rate of $R_c = \frac{7}{8}$, a BER of 10^{-4} is achieved at $SNR = 16$ dB, comparing to our *overlap* system where $T = 4$, , $R_s = 2.25$, and turbo code rate of $R_c = \frac{3}{4}$, we achieve a BER of 10^{-4} at $SNR = 9.9$ dB, which is over 6 dB improvement in SNR sustaining an increased channel throughput by a factor of 1.28. Some of the improvement may be attributed to the overlapping of the observation intervals at the receiver inherent to the *overlap* scheme. Furthermore, it is demonstrated in figure V-A and in [11], that performance improves by 2-3 dB when applying joint demodulation and decoding.

VI. CONCLUSIONS

We demonstrated by simulations that turbo codes concatenated with differential space-time modulation codes can achieve BER lower than 10^{-4} at 0.8 to 2.8 dB from code restricted capacity limit for coding rates of 0.5 to 2.25 bits per channel use over the flat fading channel, in two fading scenarios - *Overlap* and *Non-Overlap* correspondingly. Unitary code matrix differential modulation was used to exploit channel characteristics. Iterative turbo decoding and joint demodulation performs closely to capacity limit, and exhibits significant improvement over more traditional decoder without the side information feedback. This improvement is even more pronounced over short block lengths.

From the comparison in section V-D of our work to [11], it may be seen that a significant improvement in SNR is gained. This emphasizes the contribution of processing the modulation code inside the

iteration loop.

The *Non-Overlap* scheme is directly applicable to fast frequency hopping spread spectrum communication, and the *Overlap* scheme is applicable to a mobile-wireless channel, when fading dynamics is relatively slow.

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